

STEVEN PIERCE

JUNIOR SOFTWARE ENGINEER | AI-ASSISTED DEVELOPMENT | FULL-STACK & COMPUTER VISION

Ortonville, MI | 248-701-8309 | spierce34@yahoo.com | github.com/spierce34 | Software Engineering Portfolio

PROFESSIONAL SUMMARY

Junior software engineer and incoming Georgia Tech M.S. Computer Science student building production web, mobile, computer-vision, and infrastructure systems. Experienced with Python, TypeScript/JavaScript, APIs, databases, model training, and Linux operations. Uses agentic development workflows to accelerate implementation, test creation, debugging, code review, and documentation while validating changes against production systems.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL, Java, C++

Web & Mobile: React, React Native, Expo Router, Flask, FastAPI, Convex, REST APIs

AI & Computer Vision: Ultralytics YOLO, OpenCV, FFmpeg, segmentation, object detection, dataset curation, model training and validation

Data & Infrastructure: PostgreSQL/Neon, D1, R2, Cloudflare Workers/Tunnel/Load Balancing, Linux, systemd, Tailscale, restic

Engineering Workflow: Git/GitHub, CI/CD, OpenAI Codex, Claude Code, Cursor, automated testing, technical documentation

SELECTED SOFTWARE PROJECTS

PV Video Capture | AI / Computer Vision | Dec 2025 - Present Python | FastAPI | OpenCV | YOLO | PostgreSQL | Cloudflare

- Engineered multi-camera capture, low-latency inference, clipping, athlete check-in, labeling, and video-delivery workflows for real pole-vault training environments.
- Built an extraction, curation, labeling, training, and validation pipeline spanning 30,605 labeling-ready frames; trained a 397-frame runway/pit segmentation model (0.952 mask mAP50) and a 15,579-image vault-phase detector (0.947 mAP50, 0.740 mAP50-95).

Pole Vault Meet Manager | Production Platform | Aug 2025 - Present

Flask | Convex | Web | iOS | Android | Linux

- Built a production web/mobile platform that has processed 509 athlete registrations across 10 production meets, supporting check-in, scoring, organizer operations, and live results.
- Operate a two-server Linux platform at 99.90% uptime across 54K external checks since May 2026, with one-minute median incident recovery, bidirectional failover tests, and a verified 74.54 GiB disaster-recovery restore.

Landon Pole Rental | Cross-Platform App | Jun 2026 - Present

Expo Router | Cloudflare Workers | D1/R2 | Stripe

- Developed a web, iOS, and Android rental system covering inventory, checkout, agreements, role-based staff workflows, waitlists, recurring payments, and notifications.
- Implemented authorization and rental business rules with Cloudflare Workers and Better Auth; integrated D1/R2 storage, Stripe payments, Postmark email, and push notifications.

EDUCATION

Georgia Institute of Technology | *Master of Science in Computer Science*

Beginning Fall 2026

Oakland Community College | *Associate of Software Engineering*

December 2025

Relevant coursework: Computer Networks, Operating Systems, Database Management, Software Development

Eastern Michigan University | *Bachelor of Science in Biology and Chemistry*

PROFESSIONAL EXPERIENCE

365 Retail Markets | *Technical Support Specialist*

December 2025 - Present

- Own support cases in Zendesk, provide clear client updates, document root causes, identify recurring failure patterns, and escalate complex incidents through defined support paths.
- Diagnose and support Windows, Ubuntu Linux, Android/iOS, hardware, and network issues locally and through remote-support tools.

Oakland University | *Pole Vault Coach*

September 2022 - Present

- Coach NCAA Division I pole vaulters and use video analysis, performance tracking, and athlete data to guide training decisions.

Additional Experience: Waterford Mott High School - Permanent Building Substitute | March 2022 - December 2025